

# Normalizácia

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# Normalizácia

## Príklad 1

TABLE\_PRODUCT

| Product ID | Color        | Price |
|------------|--------------|-------|
| 1          | red, green   | 15.99 |
| 2          | yellow       | 23.99 |
| 3          | green        | 17.50 |
| 4          | yellow, blue | 9.99  |
| 5          | red          | 29.99 |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 1

TABLE\_PRODUCT\_PRICE

| Product ID | Price |
|------------|-------|
| 1          | 15.99 |
| 2          | 23.99 |
| 3          | 17.50 |
| 4          | 9.99  |
| 5          | 29.99 |

TABLE\_PRODUCT\_COLOR

| Product ID | Color  |
|------------|--------|
| 1          | red    |
| 1          | green  |
| 2          | yellow |
| 3          | green  |
| 4          | yellow |
| 4          | blue   |
| 5          | red    |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 2

TABLE\_PURCHASE\_DETAIL

| Customer ID | Store ID | Purchase Location |
|-------------|----------|-------------------|
| 1           | 1        | Los Angeles       |
| 1           | 3        | San Francisco     |
| 2           | 1        | Los Angeles       |
| 3           | 2        | New York          |
| 4           | 3        | San Francisco     |

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# Normalizácia

## Príklad 2

TABLE\_PURCHASE

| Customer ID | Store ID |
|-------------|----------|
| 1           | 1        |
| 1           | 3        |
| 2           | 1        |
| 3           | 2        |
| 4           | 3        |

TABLE\_STORE

| Store ID | Purchase Location |
|----------|-------------------|
| 1        | Los Angeles       |
| 2        | New Yourk         |
| 3        | San Francisco     |

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# Normalizácia

## Príklad 3

TABLE\_BOOK\_DETAIL

| Book ID | Genre ID | Genre Type | Price |
|---------|----------|------------|-------|
| 1       | 1        | Gardening  | 25,99 |
| 2       | 2        | Sports     | 14,99 |
| 3       | 1        | Gardening  | 10,99 |
| 4       | 3        | Travel     | 12,99 |
| 5       | 2        | Sports     | 17,99 |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 3

TABLE\_BOOK

| Book ID | Genre ID | Price |
|---------|----------|-------|
| 1       | 1        | 25,99 |
| 2       | 2        | 14,99 |
| 3       | 1        | 10,99 |
| 4       | 3        | 12,99 |
| 5       | 2        | 17,99 |

TABLE\_GENRE

| Genre ID | Genre Type |
|----------|------------|
| 1        | Gardening  |
| 2        | Sports     |
| 3        | Travel     |

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# Normalizácia

## Príklad 4

TABLE\_CUSTOMERS

| <b>ID</b> | <b>Name</b> | <b>Age</b> | <b>Address</b>  | <b>Orders</b>  |
|-----------|-------------|------------|-----------------|----------------|
| 100       | Sachin      | 36         | Lower West Side | Cannon XL-200  |
| 100       | Sachin      | 36         | Lower West Side | Battery XL-200 |
| 100       | Sachin      | 36         | Lower West Side | Tripod Large   |

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# Normalizácia

## Príklad 4

TABLE\_CUSTOMERS

| ID_C | Name   | Age | Address         |
|------|--------|-----|-----------------|
| 100  | Sachin | 36  | Lower West Side |

TABLE\_ORDERS

| ID | ID_C | Orders         |
|----|------|----------------|
| 10 | 100  | Cannon XL-200  |
| 11 | 100  | Battery XL-200 |
| 12 | 100  | Tripod Large   |

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## Príklad 5

| SalesStaff        |             |             |              |           |           |           |
|-------------------|-------------|-------------|--------------|-----------|-----------|-----------|
| <u>EmployeeID</u> | SalesPerson | SalesOffice | OfficeNumber | Customer1 | Customer2 | Customer3 |
| 1003              | Mary Smith  | Chicago     | 312-555-1212 | Ford      | GM        |           |
| 1004              | John Hunt   | New York    | 212-555-1212 | Dell      | HP        | Apple     |
| 1005              | Martin Hap  | Chicago     | 312-555-1212 | Boeing    |           |           |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 5

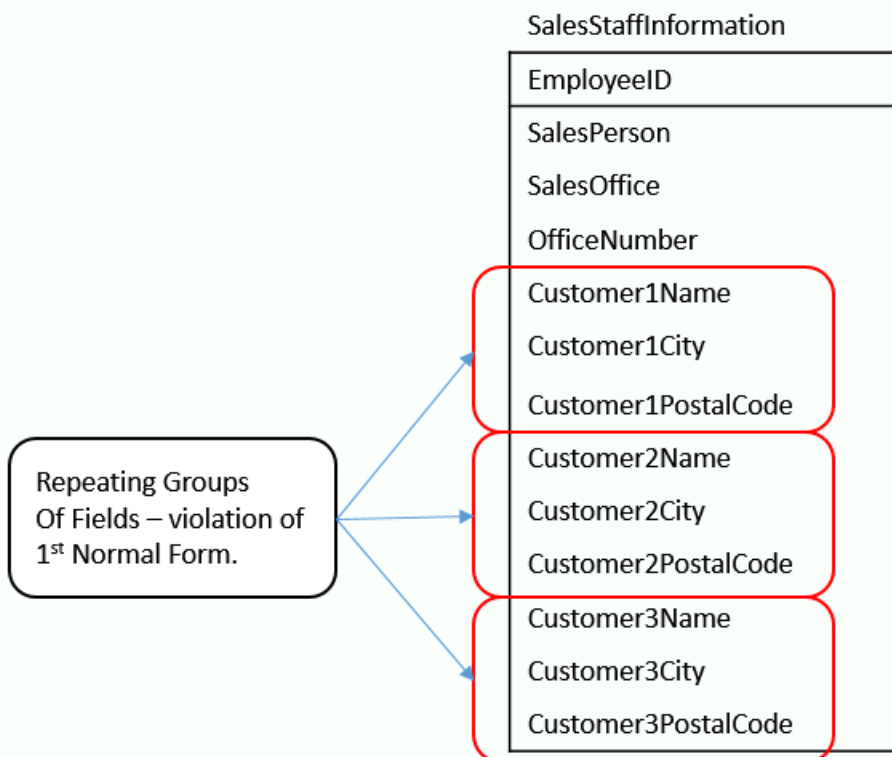
| SalesStaff        |             |             |              |           |           |           |
|-------------------|-------------|-------------|--------------|-----------|-----------|-----------|
| <u>EmployeeID</u> | SalesPerson | SalesOffice | OfficeNumber | Customer1 | Customer2 | Customer3 |
| 1003              | Mary Smith  | Chicago     | 312-555-1212 | Ford      | GM        |           |
| 1004              | John Hunt   | New York    | 212-555-1212 | Dell      | HP        | Apple     |
| 1005              | Martin Hap  | Chicago     | 312-555-1212 | Boeing    |           |           |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 5

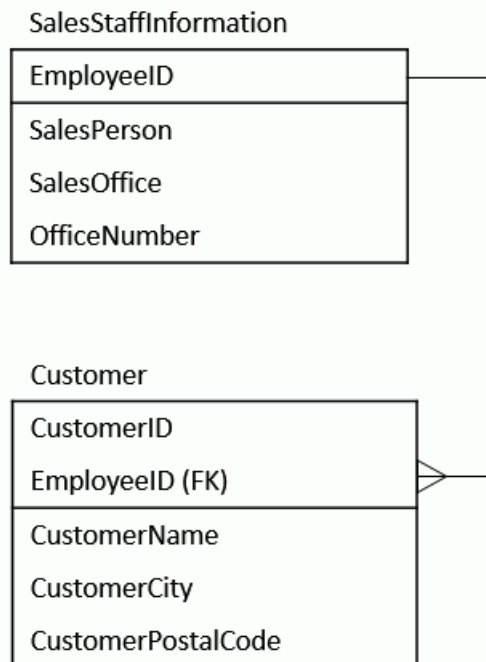
### First Normal Form Issues



# Normalizácia

## Príklad 5

### First Normal Form



# Normalizácia

## Príklad 5

| SalesStaffInformation |             |             |              |
|-----------------------|-------------|-------------|--------------|
| <u>EmployeeID</u>     | SalesPerson | SalesOffice | OfficeNumber |
| 1003                  | Mary Smith  | Chicago     | 312-555-1212 |
| 1004                  | John Hunt   | New York    | 212-555-1212 |
| 1005                  | Martin Hap  | Chicago     | 312-555-1212 |

Note: Primary Key: EmployeeID

| Customer          |            |              |              |                   |
|-------------------|------------|--------------|--------------|-------------------|
| <u>CustomerID</u> | EmployeeID | CustomerName | CustomerCity | <u>PostalCode</u> |
| C1000             | 1003       | Ford         | Dearborn     | 48123             |
| C1010             | 1003       | GM           | Detroit      | 48213             |
| C1020             | 1004       | Dell         | Austin       | 78720             |
| C1030             | 1004       | HP           | Palo Alto    | 94303             |
| C1040             | 1004       | Apple        | Cupertino    | 95014             |
| C1050             | 1005       | Boeing       | Chicago      | 60601             |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 6

TAB\_STUDENT

| ID  | NAME   | ADDRESS | SUBJECT |
|-----|--------|---------|---------|
| 100 | Adam   | Noida   | Bio     |
| 101 | Alex   | Panipat | Maths   |
| 102 | Stuart | Jammu   | Maths   |
| 103 | Adam   | Noida   | Physics |

Ktorej NF vyhovuje? Čo je PK? Ako je identifikovaný študent v tabuľke? Anomalie?

# Normalizácia

## Príklad 6

Anomália pri zmene údajov (update anomaly) :

- Pri zmene údajov študenta (napr. adresa), ktorý je v tabuľke viac krát je potrebné zmeniť údaje vo viacerých záznamoch, aby boli údaje konzistentné.
- Ako vložíme nový záznam o predmete? Nanovo vložíme zadanú adresu študenta alebo prehl'adáme tabuľku a použijeme existujúcu?
- Čo v prípade zadania predmetu Math namiesto Maths?



# Normalizácia

## Príklad 6

Anomália pri vkladaní záznamu (insert anomaly) : Ak si študent ešte nezapísal predmet, tak by neexistoval v tabuľke (neexistuje o ňom záznam ako adresa a pod.). To vedie k záznamu NULL v stĺpci SUBJECT.

Anomália pri vymazaní záznamu (drop anomaly) : Ak študent mal len jeden predmet a zruší ho, tak sa vymažú údaje o študentovi.

# Normalizácia

## Príklad 6

TAB\_STUDENT

| ID | NAME   | ADDRESS | AGE | SUBJECT      |
|----|--------|---------|-----|--------------|
| 1  | Adam   | Noida   | 15  | Bio, Physics |
| 2  | Alex   | Panipat | 14  | Maths        |
| 3  | Stuart | Jammu   | 17  | Maths        |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 6

TAB\_STUDENT

| ID_S | NAME   | ADDRESS | AGE |
|------|--------|---------|-----|
| 1    | Adam   | Noida   | 15  |
| 2    | Alex   | Panipat | 14  |
| 3    | Stuart | Jammu   | 17  |

TAB\_SUBJECTS

| ID | ID_S | SUBJECT |
|----|------|---------|
| 1  | 1    | Bio     |
| 2  | 1    | Physics |
| 3  | 2    | Maths   |
| 4  | 3    | Maths   |

Ktorej NF vyhovuje?

# Normalizácia

## Príklad 6

TAB\_STUDENT

| ID_S | NAME   | ADDRESS | AGE |
|------|--------|---------|-----|
| 1    | Adam   | Noida   | 15  |
| 2    | Alex   | Panipat | 14  |
| 3    | Stuart | Jammu   | 17  |

TAB\_SEMESTER

| ID | ID_S | ID_SUB |
|----|------|--------|
| 1  | 1    | 1      |
| 2  | 1    | 2      |
| 3  | 2    | 3      |
| 4  | 3    | 3      |

TAB\_SUBJECTS

| ID_SUB | SUBJECT |
|--------|---------|
| 1      | Bio     |
| 2      | Physics |
| 3      | Maths   |

Ktorej NF vyhovuje? Iná minimalizácia?

# Normalizácia

## Príklad 6

TAB\_STUDENT

| ID_S | NAME   | ADDRESS | AGE |
|------|--------|---------|-----|
| 1    | Adam   | Noida   | 15  |
| 2    | Alex   | Panipat | 14  |
| 3    | Stuart | Jammu   | 17  |

TAB\_SEMESTER

| ID | ID_S | ID_SUB |
|----|------|--------|
| 1  | 1    | 1      |
| 2  | 1    | 2      |
| 3  | 2    | 3      |
| 4  | 3    | 3      |

TAB\_SUBJECTS

| ID_SUB | SUBJECT |
|--------|---------|
| 1      | Bio     |
| 2      | Physics |
| 3      | Maths   |

Adresu do novej tabuľky (vid'. posledná prednáška).

# Normalizácia

## Príklad 7

### Project Management Report

**Project Code:** PC010  
**Project Title:** Pensions System

**Project Manager:** M Phillips

**Project Budget:** £24,500

| Employee No. | Employee Name | Department No. | Department Name | Hourly Rate |
|--------------|---------------|----------------|-----------------|-------------|
| S10001       | A Smith       | L004           | IT              | £22.00      |
| S10030       | L Jones       | L023           | Pensions        | £18.50      |
| S21010       | P Lewis       | L004           | IT              | £21.00      |
| S00232       | R Smith       | L003           | Programming     | £26.00      |

**Total Staff on Project:** 4

**Average Hourly Rate:** £21.88

Calculated Fields

# Normalizácia

## Príklad 7

| Project Code | Project Title   | Project Manager | Project Budget | Employee No. | Employee Name | Department No. | Department Name | Hourly Rate |
|--------------|-----------------|-----------------|----------------|--------------|---------------|----------------|-----------------|-------------|
| PC010        | Pensions System | M Phillips      | 24500          | S10001       | A Smith       | L004           | IT              | 22.00       |
| PC010        | Pensions System | M Phillips      | 24500          | S10030       | L Jones       | L023           | Pensions        | 18.50       |
| PC010        | Pensions System | M Phillips      | 24500          | S21010       | P Lewis       | L004           | IT              | 21.00       |
| PC045        | Salaries System | H Martin        | 17400          | S10010       | B Jones       | L004           | IT              | 21.75       |
| PC045        | Salaries System | H Martin        | 17400          | S10001       | A Smith       | L004           | IT              | 18.00       |
| PC045        | Salaries System | H Martin        | 17400          | S31002       | T Gilbert     | L028           | Database        | 25.50       |
| PC045        | Salaries System | H Martin        | 17400          | S13210       | W Richards    | L008           | Salary          | 17.00       |
| PC064        | HR System       | K Lewis         | 12250          | S31002       | T Gilbert     | L028           | Database        | 23.25       |
| PC064        | HR System       | K Lewis         | 12250          | S21010       | P Lewis       | L004           | IT              | 17.50       |
| PC064        | HR System       | K Lewis         | 12250          | S10034       | B James       | L009           | HR              | 16.50       |
| PC010        | Pensions System | M Phillips      | 24500          | S10001       | A Smith       | L004           | IT              | 22.00       |
|              |                 |                 |                | S10030       | L Jones       | L023           | Pensions        | 18.50       |
|              |                 |                 |                | S21010       | P Lewis       | L004           | IT              | 21.00       |
| PC045        | Salaries System | H Martin        | 17400          | S10010       | B Jones       | L004           | IT              | 21.75       |
|              |                 |                 |                | S10001       | A Smith       | L004           | IT              | 18.00       |
|              |                 |                 |                | S31002       | T Gilbert     | L028           | Database        | 25.50       |
|              |                 |                 |                | S13210       | W Richards    | L008           | Salary          | 17.00       |
| PC064        | HR System       | K Lewis         | 12250          | S31002       | T Gilbert     | L028           | Database        | 23.25       |
|              |                 |                 |                | S21010       | P Lewis       | L004           | IT              | 17.50       |
|              |                 |                 |                | S10034       | B James       | L009           | HR              | 16.50       |

# Normalizácia

## Príklad 7 (1NF)

| Project Code | Project Title   | Project Manager | Project Budget |
|--------------|-----------------|-----------------|----------------|
| PC010        | Pensions System | M Phillips      | 24500          |
| PC045        | Salaries System | H Martin        | 17400          |
| PC064        | HR System       | K Lewis         | 12250          |

| Project Code | Employee No. | Employee Name | Department No. | Department Name | Hourly Rate |
|--------------|--------------|---------------|----------------|-----------------|-------------|
| PC010        | S10001       | A Smith       | L004           | IT              | 22.00       |
| PC010        | S10030       | L Jones       | L023           | Pensions        | 18.50       |
| PC010        | S21010       | P Lewis       | L004           | IT              | 21.00       |
| PC045        | S10010       | B Jones       | L004           | IT              | 21.75       |
| PC045        | S10001       | A Smith       | L004           | IT              | 18.00       |
| PC045        | S31002       | T Gilbert     | L028           | Database        | 25.50       |
| PC045        | S13210       | W Richards    | L008           | Salary          | 17.00       |
| PC064        | S31002       | T Gilbert     | L028           | Database        | 23.25       |
| PC064        | S21010       | P Lewis       | L004           | IT              | 17.50       |
| PC064        | S10034       | B James       | L009           | HR              | 16.50       |

**1NF: Repeating Attributes Removed**



# Normalizácia

## Príklad 7 (2NF)

| Project Code | Project Title   | Project Manager | Project Budget |
|--------------|-----------------|-----------------|----------------|
| PC010        | Pensions System | M Phillips      | 24500          |
| PC045        | Salaries System | H Martin        | 17400          |
| PC064        | HR System       | K Lewis         | 12250          |

| Project Code | Employee No. | Hourly Rate |
|--------------|--------------|-------------|
| PC010        | S10001       | 22.00       |
| PC010        | S10030       | 18.50       |
| PC010        | S21010       | 21.00       |
| PC045        | S10010       | 21.75       |
| PC045        | S10001       | 18.00       |
| PC045        | S31002       | 25.50       |
| PC045        | S13210       | 17.00       |
| PC064        | S31002       | 23.25       |
| PC064        | S21010       | 17.50       |
| PC064        | S10034       | 16.50       |

| Employee No. | Employee Name | Department No. | Department Name |
|--------------|---------------|----------------|-----------------|
| S10001       | A Smith       | L004           | IT              |
| S10030       | L Jones       | L023           | Pensions        |
| S21010       | P Lewis       | L004           | IT              |
| S10010       | B Jones       | L004           | IT              |
| S31002       | T Gilbert     | L028           | Database        |
| S13210       | W Richards    | L008           | Salary          |
| S10034       | B James       | L009           | HR              |

**2NF: Partial Key Dependencies Removed**

# Normalizácia

## Príklad 7 (3NF)

### Project

| Project Code | Project Title   | Project Manager | Project Budget |
|--------------|-----------------|-----------------|----------------|
| PC010        | Pensions System | M Phillips      | 24500          |
| PC045        | Salaries System | H Martin        | 17400          |
| PC064        | HR System       | K Lewis         | 12250          |

### Project Team

| Project Code | Employee No. | Hourly Rate |
|--------------|--------------|-------------|
| PC010        | S10001       | 22.00       |
| PC010        | S10030       | 18.50       |
| PC010        | S21010       | 21.00       |
| PC045        | S10010       | 21.75       |
| PC045        | S10001       | 18.00       |
| PC045        | S31002       | 25.50       |
| PC045        | S13210       | 17.00       |
| PC064        | S31002       | 23.25       |
| PC064        | S21010       | 17.50       |
| PC064        | S10034       | 16.50       |

### Employee

| Employee No. | Employee Name | Department No. * |
|--------------|---------------|------------------|
| S10001       | A Smith       | L004             |
| S10030       | L Jones       | L023             |
| S21010       | P Lewis       | L004             |
| S10010       | B Jones       | L004             |
| S31002       | T Gilbert     | L023             |
| S13210       | W Richards    | L008             |
| S10034       | B James       | L0009            |

| Department No. | Department Name |
|----------------|-----------------|
| L004           | IT              |
| L023           | Pensions        |
| L028           | Database        |
| L008           | Salary          |
| L009           | HR              |

**Department**

**3NF: Non-Key Dependencies Removed**

# Normalizácia

## Príklad 8

Employee No: *1234*

Surname: *Bloggs* Firstname: *Joe* DOB: *23/6/53*

### Qualifications Job History

- *BSCENG BSc English JPROG Junior Programmer*
- *MSCPHY Msc Physics PROG Programmer*
- *PHDCS PhD Computer Science ANALP Analyst/Programmer*
- *PM Project Manager*

Current Department: *Engineering*

Current Salary: *£23,500*

# Normalizácia

## Príklad 8

- Vytvoriť stĺpec pre položky zo správy / formulárov (ignorujú sa vypočítavaného hodnoty)
- Naplniť tabuľku
- Identifikácia PK
- Odobratie duplicitných údajov

| Emp No | Surname | First Name | DOB     | Qual Code | Qual Desc    | Job Code | Job Desc        | Current Dept | Current Salary |
|--------|---------|------------|---------|-----------|--------------|----------|-----------------|--------------|----------------|
| 1234   | Bloggs  | Joe        | 23/6/53 | BSCENG    | BSc English  | JPROG    | Jnr Prog        | Engineering  | 23,500         |
|        |         |            |         | MSCPHY    | MSc Physics  | PROG     | Programmer      |              |                |
|        |         |            |         | PHDCS     | PhD Comp Sci | ANALP    | Anal/Prog       |              |                |
|        |         |            |         |           |              | PM       | Project Manager |              |                |
| 1332   | Jones   | Mary       | 22/6/55 | MSCPHY    | MSc Physics  | JPROG    | Jnr Prog        | R&D          | 32,450         |
|        |         |            |         | PHDCS     | PhD Comp Sci | ANALP    | Anal/Prog       |              |                |
| 4321   | Smith   | Tom        | 21/3/66 | BSCF      | BSc French   | JPROG    | Jnr Prog        | R&D          | 17,500         |

# Normalizácia

## Príklad 8 (1NF)

| Emp No | Surname | Firstname | DOB     | Current Dept | Current Salary |
|--------|---------|-----------|---------|--------------|----------------|
| 1234   | Bloggs  | Joe       | 23/6/53 | Engineering  | 23,500         |
| 1332   | Jones   | Mary      | 22/6/55 | R&D          | 32,450         |
| 4321   | Smith   | Tom       | 21/3/66 | R&D          | 17,500         |

| Emp No | Qual Code | Qual Desc    |
|--------|-----------|--------------|
| 1234   | BSCENG    | BSc English  |
| 1234   | MSCPHY    | MSc Physics  |
| 1234   | PHDCS     | PhD Comp Sci |
| 1332   | MSCPHY    | MSc Physics  |
| 1332   | PHDCS     | PhD Comp Sci |
| 4321   | BSCF      | BSc French   |

| Emp No | Job Code | Job Desc        |
|--------|----------|-----------------|
| 1234   | JPROG    | Jnr Prog        |
| 1234   | PROG     | Programmer      |
| 1234   | ANALP    | Anal/Prog       |
| 1234   | PM       | Project Manager |
| 1332   | JPROG    | Jnr Prog        |
| 1332   | ANALP    | Anal/Prog       |
| 4321   | JPROG    | Jnr Prog        |

**1NF: Repeating Attributes Removed**

# Normalizácia

## Príklad 9

Employee No: 1234

Surname: Bloggs Firstname: Joe DOB: 23/6/53

Project PM Task Hourly Rate

Finance Application Ken Jones Data Analysis £10.00

DB Design £15.00

Salary Application Fiona McDonald Data Analysis £10.00

Program Design £12.00

Programming £14.00

Pension Application Ken Jones Program Design £12.00

Program Testing £12.00

Current Department: Engineering

Current Salary: £23,500

# Normalizácia

## Príklad 9

| Emp No | Surname | First Name | DOB        | Project | Project Manager | Task        | Hourly Rate | Current Dept | Current Salary |
|--------|---------|------------|------------|---------|-----------------|-------------|-------------|--------------|----------------|
| 1234   | Bloggs  | Joe        | 23.06.1953 | Finance | Ken Jones       | DA          | £10.00      | Engineering  | 23,5           |
|        |         |            |            |         |                 | DB Design   | £15.00      |              |                |
|        |         |            |            | Salary  | Fiona McDonald  | DA          | £10.00      |              |                |
|        |         |            |            |         |                 | Prog Design | £12.00      |              |                |
|        |         |            |            |         |                 | Programming | £14.00      |              |                |
| 2345   | Jones   | Henry      | 12.09.1966 | Finance | Ken Jones       | Testing     | £10.00      | Finance      | 12,5           |
|        |         |            |            | Salary  | Fiona McDonald  | Testing     | £10.00      |              |                |

# Normalizácia

## Príklad 9 – 1. dekompozícia

| <b>Emp No</b> | <b>Surname</b> | <b>Firstname</b> | <b>DOB</b> | <b>Current<br/>Dept</b> | <b>Current<br/>Salary</b> |
|---------------|----------------|------------------|------------|-------------------------|---------------------------|
| 1234          | Bloggs         | Joe              | 23.06.1953 | Engineering             | 23,5                      |
| 2345          | Jones          | Henry            | 12.09.1966 | Finance                 | 12,5                      |



# Normalizácia

## Príklad 9 – 1. dekompozícia

| Emp No | Project | Project Manager | Task        | Hourly Rate |
|--------|---------|-----------------|-------------|-------------|
| 1234   | Finance | Ken Jones       | DA          | £10.00      |
|        |         |                 | DB Design   | £15.00      |
| 1234   | Salary  | Fiona McDonald  | DA          | £10.00      |
|        |         |                 | Prog Design | £12.00      |
|        |         |                 | Programming | £14.00      |
| 2345   | Finance | Ken Jones       | Testing     | £10.00      |
| 2345   | Salary  | Fiona McDonald  | Testing     | £10.00      |

# Normalizácia

## Príklad 9 – 2. dekompozícia

| <b>Emp No</b> | <b>Surname</b> | <b>Firstname</b> | <b>DOB</b> | <b>Current<br/>Dept</b> | <b>Current<br/>Salary</b> |
|---------------|----------------|------------------|------------|-------------------------|---------------------------|
| 1234          | Bloggs         | Joe              | 23.06.1953 | Engineering             | 23,5                      |
| 2345          | Jones          | Henry            | 12.09.1966 | Finance                 | 12,5                      |

# Normalizácia

## Príklad 9 – 2. dekompozícia

| Emp No | Project | Project Manager |
|--------|---------|-----------------|
| 1234   | Finance | Ken Jones       |
| 1234   | Salary  | Fiona McDonald  |
| 2345   | Finance | Ken Jones       |
| 2345   | Salary  | Fiona McDonald  |

| Emp No | Project | Task        | Hourly Rate |
|--------|---------|-------------|-------------|
| 1234   | Finance | DA          | £10.00      |
| 1234   | Finance | DB Design   | £15.00      |
| 1234   | Salary  | DA          | £10.00      |
| 1234   | Salary  | Prog Design | £12.00      |
| 1234   | Salary  | Programming | £14.00      |
| 2345   | Finance | Testing     | £10.00      |
| 2345   | Salary  | Testing     | £10.00      |

# Normalizácia

## Príklad 10 – posledný 😊

| emp_id | emp_nationality | emp_dept                     | dept_type | dept_no_of_emp |
|--------|-----------------|------------------------------|-----------|----------------|
| 1001   | Austrian        | Production and planning      | D001      | 200            |
| 1001   | Austrian        | stores                       | D001      | 250            |
| 1002   | American        | design and technical support | D134      | 100            |
| 1002   | American        | Purchasing department        | D134      | 600            |

Ktorá NF?

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Ktorá NF? Ako na BCNF?

# Normalizácia

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- **Functional dependencies in the table above:**  
emp\_id -> emp\_nationality  
emp\_dept -> {dept\_type, dept\_no\_of\_emp}
- **Candidate key:** {emp\_id, emp\_dept}
- The table is not in BCNF as neither emp\_id nor emp\_dept alone are keys.

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# Normalizácia

## Príklad 10 – posledný ☺

### emp\_nationality table

#### **emp\_id emp\_nationality**

|      |          |
|------|----------|
| 1001 | Austrian |
| 1002 | American |

### emp\_dept table

#### **emp\_dept**

|                              |
|------------------------------|
| Production and planning      |
| stores                       |
| design and technical support |
| Purchasing department        |

#### **dept\_type**

|      |
|------|
| D001 |
| D001 |
| D134 |
| D134 |

#### **dept\_no\_of\_emp**

|     |
|-----|
| 200 |
| 250 |
| 100 |
| 600 |



# Normalizácia

## Príklad 10 – posledný ☺

### emp\_dept\_mapping table

#### **emp\_id emp\_dept**

|      |                              |
|------|------------------------------|
| 1001 | Production and planning      |
| 1001 | stores                       |
| 1002 | design and technical support |
| 1002 | Purchasing department        |

Functional dependencies:

emp\_id -> emp\_nationality

emp\_dept -> {dept\_type, dept\_no\_of\_emp}

Candidate keys:

For first table: emp\_id

For second table: emp\_dept

For third table: {emp\_id, emp\_dept}

This is now in BCNF as in both the functional dependencies left side part is a key.